

FEM - Solution to Mechanical Equilibrium and Special Elements Sections

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Introduction

This document seeks to organize the solutions to the chapters 4 and 5 of Finite Elements solutions course at *Universidade de Brasília*.

1. Theoretical Solutions

1.1 Exercise 4.1.1

In which cases the strain tensor is symmetric?

The strain tensor is symmetric in all cases where the small displacement assumption holds, as it is defined as the symmetric part of the displacement gradient.

1.2 Exercise 4.1.2

Why is the stress tensor always symmetric?

asdf

1.3 Exercise 4.1.3

Why does the classical finite element formulation assume small deformations?

asdf

1.4 Exercise 4.1.4

Give examples of typical structures corresponding to each stress state: plane stress, plane strain, and axisymmetric conditions. Provide at least three examples for each case.

asdf

1.5 Exercise 4.2.1

Explain the role of the strain-displacement matrix B in a finite element formulation.

asdf

1.6 Exercise 4.3.1

What is an equivalent nodal-load vector, and why is it needed in the finite element method?

asdf

1.7 Exercise 4.3.2

Explain the difference between an externally applied nodal force and a consistent equivalent nodal load obtained from a distributed action.

asdf

1.8 Exercise 4.3.3

How do body forces and surface tractions enter the weak form and the element load vector?

asdf

1.9 Exercise 4.4.1

What is full and reduced integration in the computation of the stiffness matrix?

asdf

1.10 Exercise 4.4.2

What are the connectivity and location vectors of an element, and how are they used in the assembly process?

asdf

1.11 Exercise 4.4.3

What is the equilibrium residual, and why must it vanish only at free degrees of freedom?

asdf

1.12 Exercise 4.4.4

How are support reactions recovered after the displacement field has been computed?

asdf

1.13 Exercise 4.5.1

Explain the origin of shear locking and volumetric locking, and indicate in which types of problems each one is most likely to appear.

asdf

1.14 Exercise 4.5.2

Why can reduced integration alleviate locking, and what numerical risk does it introduce?

asdf

1.15 Exercise 4.5.3

What are hourglass modes, and why are they associated with underintegrated elements?

asdf

1.16 Exercise 4.5.4

Why can element distortion degrade accuracy even when the interpolation order is unchanged?

asdf

1.17 Exercise 4.5.5

What is meant by ill-conditioning of the stiffness matrix, and how can it affect the numerical solution?

asdf

1.18 Exercise 4.5.6

Distinguish between an ill-conditioned stiffness matrix and a singular stiffness matrix.

asdf

1.19 Exercise 4.5.7

Give four common modeling or formulation errors that can lead to a singular stiffness matrix.

asdf

1.20 Exercise 4.5.8

Why may excessively large penalty parameters create numerical difficulties even though they enforce constraints more strongly?

asdf

1.21 Exercise 4.6.1

List five use-cases of multifreedom constraints in finite-element modeling.

asdf

1.22 Exercise 4.6.2

Compare master-slave elimination, the penalty method, and the Lagrange multiplier method for enforcing constraints. State one advantage and one drawback of each.

asdf

1.23 Exercise 5.1.1

List four applications for each of the following element types: plate, membrane, shell, and interface.

asdf

1.24 Exercise 5.1.3

Simplify the 3D formulation for isoparametric beam elements in order to derive the formulation for an isoparametric 2D beam element.

asdf

1.25 Exercise 5.1.4

Derive the formulation for a 2D interface element.

asdf

2. Exercise 4.1.5

A flat metallic plate with length 100 mm, width 40 mm, and thickness 5 mm is subjected to a uniform tensile load. When the axial force was measured as 20 kN the axial elongation was 0.1 mm and in-plane lateral contraction 0.008 mm. Determine the corresponding constitutive matrix $\mathbf{D}$ for a plane stress 2D linear elastic model.

2.1 Solution

Firstly, we need to calculate the tensions

$$\sigma_x = \frac{F}{A} = \frac{20000 \text{ N}}{0.0002 \text{ m}^2} = 100 \text{ MPa} \quad (4.1.5.1)$$

$$\varepsilon_x = \frac{\Delta L}{L} = \frac{0.0001 \text{ m}}{0.1 \text{ m}} = 0.001 \text{ m/m} \quad (4.1.5.2)$$

It is known that $E = \frac{\sigma}{\varepsilon}$, so:

$$E = \frac{100000000}{0.001} = 100000000000 \text{ Pa} = 100 \text{ GPa} \quad (4.1.5.3)$$

Similarly

$$\varepsilon_y = \frac{\Delta W}{W} = \frac{-0.000008}{0.04} = -0.0002 \quad (4.1.5.4)$$

$$\nu = -\frac{\varepsilon_y}{\varepsilon_x} = \frac{-0.0002}{0.001} = 0.2 \quad (4.1.5.5)$$

The constitutive Matrix is then defined by

$$\mathbf{D} = \frac{E}{1 - \nu^2} \cdot \begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{pmatrix} \quad (4.1.5.6)$$

so

$$\mathbf{D} = \frac{100000000000}{1 - (0.2)^2} \cdot \begin{bmatrix} 1 & 0.2 & 0 \\ 0.2 & 1 & 0 \\ 0 & 0 & \frac{1-(0.2)}{2} \end{bmatrix} \quad (4.1.5.7)$$

$$\mathbf{D} = 104.1667 \text{ GPa} \cdot \begin{bmatrix} 1 & 0.2 & 0 \\ 0.2 & 1 & 0 \\ 0 & 0 & 0.4 \end{bmatrix} \quad (4.1.5.8)$$

$$\mathbf{D} = \begin{bmatrix} 104.1667 & 20.8333 & 0 \\ 20.8333 & 104.1667 & 0 \\ 0 & 0 & 41.6667 \end{bmatrix} \text{ GPa} \quad (4.1.5.9)$$

3. Exercise 4.1.6

A cylindrical concrete sample with height 20 cm and diameter 10 cm is tested in uniaxial compression. At a point within the elastic range, the measured values are: axial stress 20 MPa, axial shortening 0.2 mm, and radial expansion 0.015 mm. Using these measurements, determine the elastic parameters E and ν , and construct the corresponding constitutive matrix $\mathbf{D}$ for a plane stress linear¹ elastic model.

3.1 Solution

Firstly, we need to define Young's module:

$$E = \frac{\sigma}{\varepsilon} \quad (4.1.6.1)$$

$$\varepsilon = \frac{\Delta L}{L} \quad (4.1.6.2)$$

Substituting (4.1.6.2) in (4.1.6.1), we get

$$E = \frac{\sigma \cdot L}{\Delta L} = \frac{20 \text{ MPa} \cdot 20 \text{ cm}}{0.02 \text{ cm}} = 20 \text{ GPa} \quad (4.1.6.3)$$

Now, we may obtain epsilon in relation to the radius:

$$\varepsilon_z = \frac{\Delta L}{L} = -0.001 \text{ m/m} \quad (4.1.6.4)$$

$$\varepsilon_r = \frac{\Delta r}{r} = 0.0003 \text{ m/m} \quad (4.1.6.5)$$

Defining Poisson:

$$\nu = -\frac{\varepsilon_r}{\varepsilon_z} = 0.3 \quad (4.1.6.6)$$

Firstly, let's analyse the constitutive matrix for an axissimetrical element, defined as

$$\begin{bmatrix} \sigma_{rr} \\ \sigma_{zz} \\ \sigma_{\theta\theta} \\ \tau_{rz} \end{bmatrix} = \mathbf{D} \cdot \begin{bmatrix} \varepsilon_{rr} \\ \varepsilon_{zz} \\ \varepsilon_{\theta\theta} \\ \gamma_{rz} \end{bmatrix} \quad (4.1.6.7)$$

Where $\mathbf{D}$ is defined by

$$\mathbf{D} = \frac{E}{(1+\nu)(1-2\nu)} \cdot \begin{bmatrix} 1-\nu & \nu & \nu & 0 \\ \nu & 1-\nu & \nu & 0 \\ \nu & \nu & 1-\nu & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \quad (4.1.6.8)$$

$$\mathbf{D} = 38461.5385 \text{ MPa} \cdot \begin{bmatrix} 0.7 & 0.3 & 0.3 & 0 \\ 0.3 & 0.7 & 0.3 & 0 \\ 0.3 & 0.3 & 0.7 & 0 \\ 0 & 0 & 0 & 0.2 \end{bmatrix} \quad (4.1.6.9)$$

¹I've read the problem incorrectly and done also the axissimetrical development, oops!

$$= \begin{bmatrix} 26923.1 & 11538.5 & 11538.5 & 0 \\ 11538.5 & 26923.1 & 11538.5 & 0 \\ 11538.5 & 11538.5 & 26923.1 & 0 \\ 0 & 0 & 0 & 7692.3 \end{bmatrix} \text{ MPa} \quad (4.1.6.10)$$

Finally, to define the same element in a plane stress linear elastic model:

$$\mathbf{D} = \frac{E}{1-\nu^2} \begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{pmatrix} \quad (4.1.6.11)$$

$$\mathbf{D} = \frac{20000 \text{ MPa}}{1-0.3^2} \cdot \begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & 0.35 \end{bmatrix} \quad (4.1.6.12)$$

$$\mathbf{D} = \begin{bmatrix} 2.1978 & 0.6593 & 0 \\ 0.6593 & 2.1978 & 0 \\ 0 & 0 & 0.7692 \end{bmatrix} \cdot 10^4 \text{ MPa} \quad (4.1.6.13)$$

4. Exercise 4.1.7

A long thick-walled cylindrical pipe with inner radius $a = 50$ mm and outer radius $b = 100$ mm is subjected to a uniform internal pressure $p = 10$ MPa. The material is linear elastic and isotropic, with Young's modulus $E = 200$ GPa and Poisson's ratio $\nu = 0.2$. Assume that the pipe is sufficiently long so that the axial strain can be neglected, i.e. $\epsilon_z = 0$. At a point located at $r = 75$ mm within the wall, the stress state is characterized by normal stresses only, with no shear components. Determine the stress components at this point and the corresponding strain components.

4.1 Solution

Since the internal pressure acts in the radial direction, and assuming axial strength can be neglected, it is valid to assume that the problem constitutes an Axisymmetric Stress, defined by

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{rr} & 0 & \sigma_{rz} \\ 0 & \sigma_{\theta\theta} & 0 \\ \sigma_{zr} & 0 & \sigma_{zz} \end{bmatrix} \quad (4.1.7.1)$$

It is known, for the current material, that

$$\nu = 0.2 \quad (4.1.7.2)$$

$$E = 200 \text{ MPa} \quad (4.1.7.3)$$

Thus, for the Axisymmetric condition:

$$\begin{bmatrix} \sigma_{rr} \\ \sigma_{zz} \\ \sigma_{\theta\theta} \\ \tau_{rz} \end{bmatrix} = \mathbf{D} \cdot \begin{bmatrix} \epsilon_{rr} \\ \epsilon_{zz} \\ \epsilon_{\theta\theta} \\ \gamma_{rz} \end{bmatrix} \quad (4.1.7.4)$$

But, since axial strain can be neglected, that is, $\epsilon_{zz} = 0$, we can define it as

$$\begin{bmatrix} \sigma_{rr} \\ \sigma_{\theta\theta} \\ \tau_{rz} \end{bmatrix} = \mathbf{D} \cdot \begin{bmatrix} \epsilon_{rr} \\ \epsilon_{\theta\theta} \\ \gamma_{rz} \end{bmatrix} \quad (4.1.7.5)$$

Where $\mathbf{D}$, in this specific case, will be

$$\mathbf{D} = \frac{E}{(1+\nu)(1-2\nu)} \cdot \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \quad (4.1.7.6)$$

$$\mathbf{D} = \frac{200}{(1+0.2)(1-0.4)} \cdot \begin{bmatrix} 0.8 & 0.2 & 0 \\ 0.2 & 0.8 & 0 \\ 0 & 0 & 0.3 \end{bmatrix} \quad (4.1.7.7)$$

$$\mathbf{D} = \begin{bmatrix} 222.2222 & 55.5556 & 0 \\ 55.5556 & 222.2222 & 0 \\ 0 & 0 & 83.3333 \end{bmatrix} \text{ MPa} \quad (4.1.7.8)$$

With the defined matrix:²

²This problem is incomplete

5. Exercise 4.2.2

It is known that

$$u = \sum_{i=1}^3 N_i(\xi) u_i = \mathbf{N} \mathbf{U}^e \quad (4.1.7.1)$$

$$\varepsilon = \frac{du}{dx} = \frac{d\mathbf{N}^T}{dx} \mathbf{U}^e \quad (4.1.7.2)$$

$$\frac{d\mathbf{N}^T}{dx} = \frac{d\mathbf{N}^T}{d\xi} \cdot J^{-1} \quad (4.1.7.3)$$

$$\varepsilon = \frac{du}{dx} = \frac{1}{J} \frac{d\mathbf{N}^T}{d\xi} \mathbf{U}^e \quad (4.1.7.4)$$

Thus

$$\begin{aligned} \mathbf{B} &= \frac{1}{J} \frac{d\mathbf{N}^T}{d\xi} \\ &= \frac{1}{J} \left(\frac{dN_1}{d\xi} \quad \frac{dN_2}{d\xi} \quad \frac{dN_3}{d\xi} \right) \end{aligned} \quad (4.1.7.5)$$

For a 3-node bar element

$$\mathbf{N}^T = \left(\frac{\xi(\xi-1)}{2} \quad \frac{\xi(\xi+1)}{2} \quad 1 - \xi^2 \right) \quad (4.1.7.6)$$

$$\frac{d\mathbf{N}^T}{d\xi} = \left(\frac{2\xi-1}{2} \quad \frac{2\xi+1}{2} \quad -2\xi \right) \quad (4.1.7.7)$$

$$J = \mathbf{X}^T \frac{d\mathbf{N}}{d\xi} \quad (4.1.7.8)$$

Thus

$$\mathbf{B} = \left(\mathbf{X}^T \frac{d\mathbf{N}}{d\xi} \right)^{-1} \frac{d\mathbf{N}^T}{d\xi} \quad (4.1.7.9)$$

For

$$\mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad (4.1.7.10)$$

6. Exercise 4.2.3

Find the matrix $\mathbf{B}$ for a three-node triangular element with nodal coordinates (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) . Why is the matrix $\mathbf{B}$ constant along the element, and what does that imply?

6.1 Solution

First of all, a 3 node triangular element is defined by the following shape functions

$$\mathbf{N} = \begin{pmatrix} 1 - \xi - \eta \\ \xi \\ \eta \end{pmatrix} \quad (4.2.3.1)$$

For

$$\mathbf{x} = \sum_{i=1}^3 N_i(\xi, \eta) \cdot \mathbf{x}_i \quad (4.2.3.2)$$

$$\mathbf{u} = \sum_{i=1}^3 N_i(\xi, \eta) \cdot \mathbf{u}_i \quad (4.2.3.3)$$

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix} = \mathbf{B} \cdot \begin{bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \end{bmatrix} \quad (4.2.3.4)$$

$$\mathbf{B} = (\mathbf{B}_1 \ \mathbf{B}_2 \ \mathbf{B}_3) \quad (4.2.3.5)$$

$$\mathbf{B}_i = \begin{pmatrix} \frac{\partial N_i}{\partial x} & 0 \\ 0 & \frac{\partial N_i}{\partial y} \\ \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} \end{pmatrix} \quad (4.2.3.6)$$

It is known that

$$\frac{\partial N_i}{\partial x} = \frac{\partial N_i}{\partial \xi} \mathbf{J}^{-1} \quad (4.2.3.7)$$

$$x(\xi, \eta) = (1 - \xi - \eta)x_1 + \xi x_2 + \eta x_3 = x_1 + \xi(x_2 - x_1) + \eta(x_3 - x_1) \quad (4.2.3.8)$$

$$y(\xi, \eta) = (1 - \xi - \eta)y_1 + \xi y_2 + \eta y_3 = y_1 + \xi(y_2 - y_1) + \eta(y_3 - y_1) \quad (4.2.3.9)$$

Thus the Jacobian is

$$\mathbf{J} = \begin{pmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{pmatrix} = \begin{pmatrix} x_{21} & y_{21} \\ x_{31} & y_{31} \end{pmatrix} \quad (4.2.3.10)$$

$$J = \det(\mathbf{J}) = x_{21}y_{31} - x_{31}y_{21} = (x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1) \quad (4.2.3.11)$$

Which, as seen by the shoelace formula, $J = 2A$.

It is known that

$$\mathbf{J}^{-1} = \frac{\text{adj}(\mathbf{J})}{\det(\mathbf{J})} = \frac{1}{2A} \begin{pmatrix} y_{31} & -y_{21} \\ -x_{31} & x_{21} \end{pmatrix} \quad (4.2.3.12)$$

For the first node

$$\begin{pmatrix} \frac{\partial N_1}{\partial x} \\ \frac{\partial N_1}{\partial y} \end{pmatrix} = \mathbf{J}^{-1} \begin{pmatrix} \frac{\partial N_1}{\partial \xi} \\ \frac{\partial N_1}{\partial \eta} \end{pmatrix} = \frac{1}{2A} \begin{pmatrix} y_{31} & -y_{21} \\ -x_{31} & x_{21} \end{pmatrix} \begin{pmatrix} -1 \\ -1 \end{pmatrix} \quad (4.2.3.13)$$

$$\begin{pmatrix} \frac{\partial N_1}{\partial x} \\ \frac{\partial N_1}{\partial y} \end{pmatrix} = \frac{1}{2A} \begin{pmatrix} y_{21} - y_{31} \\ x_{31} - x_{21} \end{pmatrix} = \frac{1}{2A} \begin{pmatrix} y_{23} \\ x_{32} \end{pmatrix} \quad (4.2.3.14)$$

For the second node

$$\begin{pmatrix} \frac{\partial N_2}{\partial x} \\ \frac{\partial N_2}{\partial y} \end{pmatrix} = \mathbf{J}^{-1} \begin{pmatrix} \frac{\partial N_2}{\partial \xi} \\ \frac{\partial N_2}{\partial \eta} \end{pmatrix} = \frac{1}{2A} \begin{pmatrix} y_{31} & -y_{21} \\ -x_{31} & x_{21} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (4.2.3.15)$$

$$\begin{pmatrix} \frac{\partial N_2}{\partial x} \\ \frac{\partial N_2}{\partial y} \end{pmatrix} = \frac{1}{2A} \begin{pmatrix} y_{31} \\ x_{13} \end{pmatrix} \quad (4.2.3.16)$$

For the third node

$$\begin{pmatrix} \frac{\partial N_3}{\partial x} \\ \frac{\partial N_3}{\partial y} \end{pmatrix} = \frac{1}{2A} \begin{pmatrix} y_{12} \\ x_{21} \end{pmatrix} \quad (4.2.3.17)$$

Thus, finally:

$$\mathbf{B} = \frac{1}{2A} \cdot \begin{pmatrix} y_{23} & 0 & y_{31} & 0 & y_{12} & 0 \\ 0 & x_{32} & 0 & x_{13} & 0 & x_{21} \\ x_{32} & y_{23} & x_{13} & y_{31} & x_{21} & y_{12} \end{pmatrix} \quad (4.2.3.18)$$

Note the notation $x_{ij} = x_i - x_j$.

The strain-displacement matrix $\mathbf{B}$ is uniform across the entire domain of the element Ω_e :

$$\mathbf{B}(x, y) = \text{const} \quad \forall (x, y) \in \Omega_e \quad (4.2.3.19)$$

This characteristic is true because the Jacobian $\mathbf{J}$ is constant, because the element is linear.

7. Exercise 4.2.4

For the quadrilateral element shown below, and given its nodal coordinates and nodal-displacement vector

$$C = \begin{pmatrix} 0 & 0 \\ 4 & 2 \\ 4 & 4 \\ 0 & 2 \end{pmatrix} \text{ m} \quad \text{and} \quad U^{(e)} = \begin{pmatrix} 0 \\ 0 \\ 0.01 \\ 0.01 \\ 0.015 \\ 0.015 \\ 0 \\ 0.015 \end{pmatrix} \text{ m} \quad (4.2.4.1)$$

as shown in the figure. Approximate the strain vector at the natural coordinates $(\xi, \eta) = \left(-\sqrt{\frac{1}{3}}, \sqrt{\frac{1}{3}}\right)$.

7.1 Solution

First step, is to define the shape functions and its Jacobian

$$N(\xi, \eta) = \begin{pmatrix} \frac{(1-\xi)(1-\eta)}{4} \\ \frac{(1+\xi)(1-\eta)}{4} \\ \frac{(1+\xi)(1+\eta)}{4} \\ \frac{(1-\xi)(1+\eta)}{4} \end{pmatrix} \quad (4.2.4.2)$$

Or simply

$$N_i(\xi, \eta) = \frac{1}{4}(\xi_i - \xi)(\eta_i - \eta) \quad (4.2.4.3)$$

Taking its derivatives

$$\begin{pmatrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \end{pmatrix} = \begin{pmatrix} \eta - \eta_i \\ \xi - \xi_i \end{pmatrix} \quad (4.2.4.4)$$

$$\begin{pmatrix} x(\xi, \eta) \\ y(\xi, \eta) \end{pmatrix} = \sum_{i=1}^n \begin{pmatrix} \frac{x_i}{4}(\xi_i - \xi)(\eta_i - \eta) \\ \frac{y_i}{4}(\xi_i - \xi)(\eta_i - \eta) \end{pmatrix} \quad (4.2.4.5)$$

The strain vector for a two dimensional element, which can suffer strains in xx and yy, and the shear strain xy, is defined by

$$\varepsilon = (B_1 \ B_2 \ B_3 \ B_4) \cdot U^e \quad (4.2.4.6)$$

Since we already know U^e , we just have to use C to define B_i :

$$B_i = \begin{pmatrix} \frac{\partial N_i}{\partial x} & 0 \\ 0 & \frac{\partial N_i}{\partial y} \\ \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} \end{pmatrix} \quad (4.2.4.7)$$

$$\frac{\partial N_i}{\partial x} = \frac{\partial N_i}{\partial \xi} \mathbf{J}^{-1} \quad (4.2.4.8)$$

For $(\xi, \eta) = \left(-\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right)$, we have:

$$\frac{\partial \mathbf{N}}{\partial \xi} = \begin{bmatrix} -0.1057 & -0.3943 \\ 0.1057 & -0.1057 \\ 0.3943 & 0.1057 \\ -0.3943 & 0.3943 \end{bmatrix} \quad (4.2.4.9)$$

And the Jacobian is:

$$\mathbf{J} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \quad (4.2.4.10)$$

Thus its inverse is:

$$\mathbf{J}^{-1} = \begin{bmatrix} 0.5 & 0 \\ -0.5 & 1 \end{bmatrix} \quad (4.2.4.11)$$

With (4.2.4.9) and (4.2.4.11) defined, it is simple to obtain the $\mathbf{B}$:

$$\mathbf{B} = \begin{bmatrix} 0.1443 & 0 & 0.1057 & 0 & 0.1443 & 0 & -0.3943 & 0 \\ 0 & -0.3943 & 0 & -0.1057 & 0 & 0.1057 & 0 & 0.3943 \\ -0.3943 & 0.1443 & -0.1057 & 0.1057 & 0.1057 & 0.1443 & 0.3943 & -0.3943 \end{bmatrix} \quad (4.2.4.12)$$

Substituting this into (4.2.4.6), with the use of Julia, we obtain the resultant ε matrix:

```
using LinearAlgebra

C = [
    0 0;
    4 2;
    4 4;
    0 2;
]

U = [
    0;
    0;
    0.01;
    0.01;
    0.015;
    0.015;
    0;
    0.015;
]

function N_derivs(xi, eta)
    return [
        0.25*(-1.0 + eta)  0.25*(-1.0 + xi)
        0.25*(+1.0 - eta)  0.25*(-1.0 - xi)
        0.25*(+1.0 + eta)  0.25*(+1.0 + xi)
        0.25*(-1.0 - eta)  0.25*(+1.0 - xi)
    ]
end

function Jacobian(C, xi, eta)
    # Get the 4x2 matrix of shape function derivatives
    dNdξ = N_derivs(xi, eta)
```

```

        return C' * dNdξ
    end

    function getB(C, xi, eta)
        dNdx = N_derivs(xi, eta) * inv(Jacobian(C, xi, eta))
        B = nothing
        for row in eachrow(dNdx)
            dx, dy = row
            Bi = [
                dx 0;
                0 dy;
                dy dx;
            ]
            if B === nothing
                B = Bi
            else
                B = [B Bi]
            end
        end

        return B
    end

    B = getB(C, -sqrt(1/3), sqrt(1/3))

    ε = B * U

    ε

```

```

3-element Vector{Float64}:
 0.003221687836487032
 0.006443375672974064
-0.0021650635094610966

```

Hence

$$\varepsilon = \begin{bmatrix} 0.003221688 \\ 0.006443376 \\ -0.002165064 \end{bmatrix} \quad (4.2.4.13)$$

8. Exercise 4.2.5

Find an expression for the matrix B of a three-node bar element in 2D as a function of the Jacobian and the local coordinate ξ . Consider $\varepsilon = \frac{\partial u}{\partial s}$, where s is the curvilinear coordinate along the bar path.

8.1 Solution

Strain is defined as the derivative of displacement, that is:

$$\varepsilon = \frac{\partial u}{\partial s} \quad (4.2.5.1)$$

We know that

$$J = \sqrt{\left(\frac{\partial x}{\partial \xi}\right)^2 + \left(\frac{\partial y}{\partial \xi}\right)^2} = \frac{\partial s}{\partial \xi} \quad (4.2.5.2)$$

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial s} = \frac{1}{J} \frac{\partial u}{\partial \xi} \quad (4.2.5.3)$$

It is simple to denote that

$$\frac{\partial u}{\partial \xi} = \frac{\partial N}{\partial \xi} U^e \quad (4.2.5.4)$$

Where, in this case:

$$\begin{aligned} \frac{\partial N}{\partial \xi} &= \begin{pmatrix} \frac{\partial N_1}{\partial \xi} & 0 & \frac{\partial N_2}{\partial \xi} & 0 & \frac{\partial N_2}{\partial \xi} & 0 \\ 0 & \frac{\partial N_1}{\partial \xi} & 0 & \frac{\partial N_2}{\partial \xi} & 0 & \frac{\partial N_2}{\partial \xi} \end{pmatrix} \\ U^e &= \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \end{pmatrix} \end{aligned} \quad (4.2.5.5)$$

Substituting it into (4.2.5.3):

$$\varepsilon = \frac{1}{J} \frac{\partial N}{\partial \xi} U^e \quad (4.2.5.6)$$

To obtain the strain along the element we can simply project it tangentially:

$$\hat{\mathbf{r}}^T = \frac{\mathbf{J}^T}{J} = (c \ s) \quad (4.2.5.7)$$

$$\varepsilon = \frac{\hat{\mathbf{r}}^T}{J} \frac{\partial N}{\partial \xi} U^e = \frac{\mathbf{J}^T}{J^2} \frac{\partial N}{\partial \xi} U^e = \mathbf{B}(\xi) U^e \quad (4.2.5.8)$$

$$\mathbf{B}(\xi) = \frac{\mathbf{J}}{J^2} \frac{\partial N}{\partial \xi} \quad (4.2.5.9)$$

9. Exercise 4.3.4

For the elements below, determine the nodal forces equivalent to body forces. Consider the element area A , thickness t , and unit weight γ .

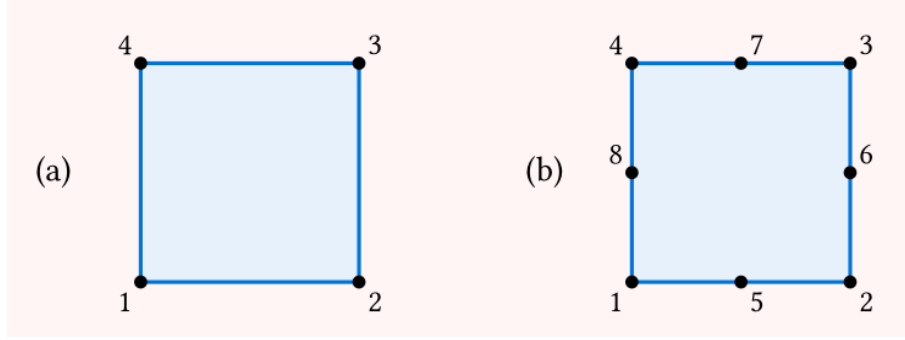


Figure 9.1: 4-node (a) and 8-node (b) Square Elements

9.1 Solution

The equivalent vertical nodal-force vector is

$$\mathbf{F}_v = t \int_{A'} \mathbf{N}^T b_v J d\eta d\xi \quad (4.3.4.1)$$

Since the Area is A , the Jacobian will be

$$J = \frac{A}{4} \quad (4.3.4.2)$$

9.1.1 (a)

The shape function for a 4-node square element will be

$$N_i = \frac{1}{4}(1 + \xi_i \xi)(1 + \eta_i \eta) \quad (4.3.4.3)$$

Thus

$$\mathbf{F}_v = -\frac{\gamma t A}{16} \int_{-1}^1 \int_{-1}^1 \begin{pmatrix} (1 - \xi)(1 - \eta) \\ (1 + \xi)(1 - \eta) \\ (1 + \xi)(1 + \eta) \\ (1 - \xi)(1 + \eta) \end{pmatrix} d\xi d\eta \quad (4.3.4.4)$$

Note that

$$\int_{-1}^1 (1 \pm x) dx = 2 \quad (4.3.4.5)$$

$$\mathbf{F}_v = -\frac{\gamma t A}{16} \begin{pmatrix} 4 \\ 4 \\ 4 \\ 4 \end{pmatrix} = -\frac{\gamma t A}{4} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad (4.3.4.6)$$

9.1.2 (b)

Similarly, for the 8-node element,

$$\mathbf{F}_v = -\frac{\gamma t A}{4} \int_{A'} \mathbf{N}^T d\eta d\xi \quad (4.3.4.7)$$

$$\mathbf{N}^T = \begin{pmatrix} N_1 \\ N_2 \\ N_3 \\ N_4 \\ N_5 \\ N_6 \\ N_7 \\ N_8 \end{pmatrix} = \begin{pmatrix} \frac{1}{4}(1-\xi)(1-\eta)(-1-\xi-\eta) \\ \frac{1}{4}(1+\xi)(1-\eta)(-1+\xi-\eta) \\ \frac{1}{4}(1+\xi)(1+\eta)(-1+\xi+\eta) \\ \frac{1}{4}(1-\xi)(1+\eta)(-1-\xi+\eta) \\ \frac{1}{2}(1-\eta)(1-\xi^2) \\ \frac{1}{2}(1+\xi)(1-\eta^2) \\ \frac{1}{2}(1+\eta)(1-\xi^2) \\ \frac{1}{2}(1-\xi)(1-\eta^2) \end{pmatrix} \quad (4.3.4.8)$$

Solving it in julia, we obtain

```
using LinearAlgebra

function Nvector(xi, eta)
    return [
        1/4 * (1 - xi) * (1 - eta) * (-1 - xi - eta);
        1/4 * (1 + xi) * (1 - eta) * (-1 + xi - eta);
        1/4 * (1 + xi) * (1 + eta) * (-1 + xi + eta);
        1/4 * (1 - xi) * (1 + eta) * (-1 - xi + eta);
        1/2 * (1 - eta) * (1 - xi^2);
        1/2 * (1 + xi) * (1 - eta^2);
        1/2 * (1 + eta) * (1 - xi^2);
        1/2 * (1 - xi) * (1 - eta^2);
    ]
end

quad_2 = [
    (-1 / sqrt(3), 1.0),
    ( 1 / sqrt(3), 1.0)
]

function obtain_integral(Nvector)
    # Initialize result based on the actual length of the vector (8 elements)
    result = zeros(length(Nvector(0.0, 0.0)))

    for i in eachindex(result)
        integral = 0.0
        for (xi, wi) in quad_2
            for (eta, wj) in quad_2
                # Correctly indexing the i-th shape function
                integral += Nvector(xi, eta)[i] * wi * wj
            end
        end
    end
end
```

```

        result[i] = integral
    end

    return result
end

display(obtain_integral(Nvector))

```

```

8-element Vector{Float64}:
-0.3333333333333331
-0.3333333333333331
-0.3333333333333331
-0.3333333333333331
 1.3333333333333333
 1.3333333333333333
 1.3333333333333333
 1.333333333333328

```

$$\int_{A'} \mathbf{N}^T d\xi d\eta = [-0.3333 \ -0.3333 \ -0.3333 \ -0.3333 \ 1.3333 \ 1.3333 \ 1.3333 \ 1.3333] \quad (4.3.4.9)$$

Finally

$$\mathbf{F}_v = \frac{\gamma t A}{12} [1 \ 1 \ 1 \ 1 \ -4 \ -4 \ -4 \ -4] \quad (4.3.4.10)$$

10. Exercise 4.3.5

Determine the equivalent nodal forces corresponding to the distributed load per unit length q .

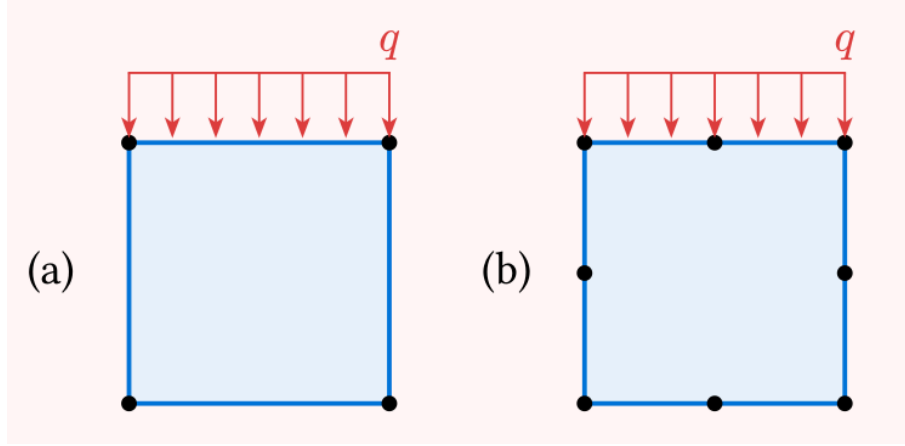


Figure 10.1: distributed vertical load q applied to a 4-node (a) and 8-node (b) square element

10.1 Solution

10.1.1 (a)

For a distributed load q acting on an element edge, the equivalent nodal forces are obtained by integrating the traction over the 1D boundary (the edge) rather than the 2D area:

$$\mathbf{F}_t = \int_L \mathbf{N}^T \mathbf{t} dL = \int_{-1}^1 \mathbf{N}^T \mathbf{t} J d\xi \quad (4.3.5.1)$$

The load is a vertical traction per unit length, so the traction vector is simply $\mathbf{t} = \begin{pmatrix} 0 \\ -q \end{pmatrix}$.

Since the integration is along a 1D edge of length L , the corresponding line Jacobian is:

$$J = \frac{L}{2} \quad (4.3.5.2)$$

Assuming the load is applied to the top edge connecting nodes 4 and 3, we evaluate the shape functions at $\eta = 1$. The relevant 2D shape functions simplify to 1D linear shape functions along ξ :

$$\begin{aligned} N_4(\xi, 1) &= \frac{1}{2}(1 - \xi) \\ N_3(\xi, 1) &= \frac{1}{2}(1 + \xi) \end{aligned} \quad (4.3.5.3)$$

The vertical nodal forces for these two nodes are evaluated as:

$$\begin{pmatrix} F_{4y} \\ F_{3y} \end{pmatrix} = \int_{-1}^1 \begin{pmatrix} N_4 \\ N_3 \end{pmatrix} (-q) \left(\frac{L}{2} \right) d\xi \quad (4.3.5.4)$$

$$\begin{pmatrix} F_{4y} \\ F_{3y} \end{pmatrix} = -\frac{qL}{2} \int_{-1}^1 \begin{pmatrix} \frac{1}{2}(1 - \xi) \\ \frac{1}{2}(1 + \xi) \end{pmatrix} d\xi \quad (4.3.5.5)$$

Because the integral of each 1D linear shape function from -1 to 1 evaluates exactly to 1 , we obtain the vertical equivalent forces:

$$\begin{pmatrix} F_{4y} \\ F_{3y} \end{pmatrix} = \frac{qL}{2} \begin{pmatrix} -1 \\ -1 \end{pmatrix} \quad (4.3.5.6)$$

10.1.2 (b)

For the same distributed load q acting now on an 8-node element, the shape functions are that of a 8-node square element, evaluated at $\eta = 1$

$$\mathbf{N}^T = \begin{pmatrix} N_3 \\ N_4 \\ N_7 \end{pmatrix} = \begin{pmatrix} \frac{\xi}{2}(\xi + 1) \\ \frac{\xi}{2}(\xi - 1) \\ 1 - \xi^2 \end{pmatrix} \quad (4.3.5.7)$$

$$\begin{aligned} \begin{pmatrix} F_{4y} \\ F_{7y} \\ F_{3y} \end{pmatrix} &= \int_{-1}^1 \begin{pmatrix} N_4 \\ N_7 \\ N_3 \end{pmatrix} (-q) \left(\frac{L}{2} \right) d\xi = \\ &= -\frac{qL}{2} \int_{-1}^1 \begin{pmatrix} \frac{\xi}{2}(\xi + 1) \\ \frac{\xi}{2}(\xi - 1) \\ 1 - \xi^2 \end{pmatrix} d\xi \end{aligned} \quad (4.3.5.8)$$

This equates to

$$\begin{pmatrix} F_{4y} \\ F_{7y} \\ F_{3y} \end{pmatrix} = -\frac{qL}{6} \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} \quad (4.3.5.9)$$

11. Exercise 4.3.6

The nodal coordinates of a 3-node bar element are $(x_1, y_1) = (-1, -1)$, $(x_2, y_2) = (1, 1)$, and $(x_3, y_3) = (0, 1)$. If the unit weight is $\gamma = 70\text{kN/m}^3$ and the area is $A = 0.02\text{m}^2$, calculate the equivalent nodal forces due to the element weight using numerical integration.

11.1 Solution

The following script will solve the problem stated

```
using LinearAlgebra

X = [
    -1.0 -1.0;
     1.0  1.0;
     0.0  1.0
]

γ = 70.0 # kN/m³
A = 0.02 # m²

function body_force()
    return [0.0, -γ * A]
end

function N(xi)
    return [
        0.5 * xi * (xi - 1.0);
        0.5 * xi * (xi + 1.0);
        1.0 - xi²
    ]
end

function partial_N(xi)
    return [
        xi - 0.5;
        xi + 0.5;
        -2.0 * xi
    ]
end

function Jacobian(X, xi)
    return X' * partial_N(xi)
end

function Fb(N, b, J, X)
    quad_3 = [
        (-sqrt(3/5), 5/9),
        ( 0.0,      8/9),
        ( sqrt(3/5), 5/9)
    ]

    result = zeros(3, 2)
```

```
    for (xi, w) in quad_3
        result += N(xi) * transpose(b()) * norm(J(X, xi)) * w
    end

    return result
end

Fb(N, body_force, Jacobian, X)
```

```
3×2 Matrix{Float64}:
 0.0  -1.38635
 0.0  -0.423948
 0.0  -2.96677
```

12. Exercise 4.3.7

Find the nodal forces corresponding to horizontal and normal tractions with $q = 1 \text{ kN/m}$ as shown below. Assume the element thickness is $h = 1 \text{ m}$. The coordinates matrix is:

$$\mathbf{X} = \begin{pmatrix} 0 & 0 \\ 4 & 0 \\ 4 & 4 \\ 0 & 4 \\ 2 & 0 \\ 3 & 2 \\ 2 & 4 \\ 0 & 2 \end{pmatrix} \text{ m} \quad (4.3.7.1)$$

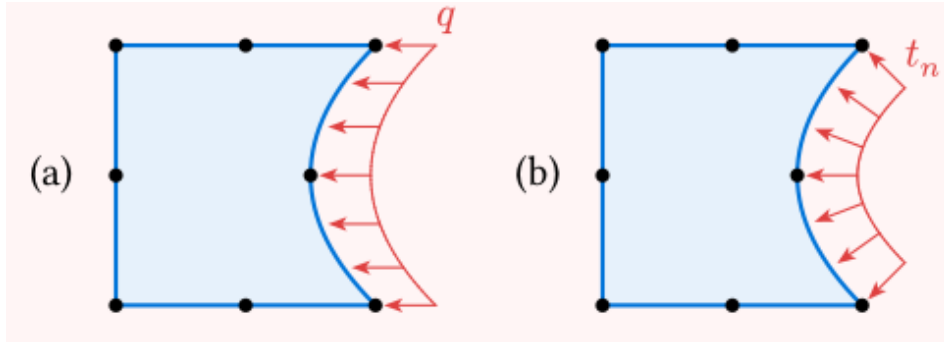


Figure 12.1: 8-node irregular element with horizontal (a) and normal (b) tractions

12.1 Solution

The equivalent nodal forces are

$$\mathbf{F}_t = \int_A \mathbf{N}^T \mathbf{t} J d\xi d\eta \quad (4.3.7.2)$$

The shape functions for the nodes of interest are defined as

$$\begin{pmatrix} N_2 \\ N_3 \\ N_6 \end{pmatrix} = \begin{pmatrix} \frac{\xi}{2}(\xi - 1) \\ \frac{\xi}{2}(\xi + 1) \\ 1 - \xi^2 \end{pmatrix} \quad (4.3.7.3)$$

In the integral being

$$\mathbf{N}^T = \begin{pmatrix} N_2 & 0 \\ 0 & N_2 \\ N_3 & 0 \\ 0 & N_3 \\ N_6 & 0 \\ 0 & N_6 \end{pmatrix} \quad (4.3.7.4)$$

The Jacobian depends on the coordinate matrix $\mathbf{X}$ and the current parametric coordinate ξ for the shape function:

$$J(\xi) = \mathbf{X}^T \frac{d\mathbf{N}(\xi)}{d\xi} \quad (4.3.7.5)$$

12.1.1 (a)

The traction vector is then defined as

$$\mathbf{t} = \begin{pmatrix} -q \\ 0 \end{pmatrix} \quad (4.3.7.6)$$

Then we must define

$$\mathbf{F}_t = \int_{-1}^1 \begin{pmatrix} N_2 & 0 \\ 0 & N_2 \\ N_3 & 0 \\ 0 & N_3 \\ N_6 & 0 \\ 0 & N_6 \end{pmatrix} \begin{pmatrix} -q \\ 0 \end{pmatrix} \|J\| d\xi \quad (4.3.7.7)$$

For the x coordinate, equates to

$$F_{xt} = -q \int_{-1}^1 \begin{pmatrix} N_2 \\ N_3 \\ N_6 \end{pmatrix} \|J\| d\xi \quad (4.3.7.8)$$

To solve this in julia:

```
using LinearAlgebra

function main()

    X = [
        4 0;
        4 4;
        3 2;
    ]

    dNdx(xi) = [
        xi - 1/2;
        xi + 1/2;
        -2 * xi
    ]

    # define quad_2 values
    quad_2 = [
        (-1 / sqrt(3), 1.0),
        (1 / sqrt(3), 1.0)
    ]

    # shape functions as vector
    N(xi) = [
        xi / 2 * (xi - 1);
        xi / 2 * (xi + 1);
        1 - xi^2
    ]
end
```

```

]

result = zeros(3)

for (xi, w) in quad_2
    result += N(xi) * norm(X' * dNdxi(xi)) * w
end

return result
end

main()

```

```

3-element Vector{Float64}:
 0.7698003589195012
 0.7698003589195008
 3.0792014356780024

```

Thus

$$F_{xt} = \begin{pmatrix} F_{2xt} \\ F_{3xt} \\ F_{6xt} \end{pmatrix} = -q \begin{bmatrix} 0.7698 \\ 0.7698 \\ 3.0792 \end{bmatrix} \quad (4.3.7.9)$$

12.2 (b)

In relation to the last problem, the only difference is defined in the tension vector, which is now:

$$t(\xi) = -\hat{n}(\xi)q \quad (4.3.7.10)$$

We can define the normal vector with the Jacobian:

$$\hat{n}(\xi) \perp \hat{J}(\xi) = \frac{\mathbf{J}}{\|\mathbf{J}\|} \quad (4.3.7.11)$$

For a 2D Jacobian vector, then:

$$\hat{n} = (\hat{J}_2 \ -\hat{J}_1)^T \quad (4.3.7.12)$$

So, simply:

$$\mathbf{F}_t = -q \int_{-1}^1 \begin{pmatrix} N_2 & 0 \\ 0 & N_2 \\ N_3 & 0 \\ 0 & N_3 \\ N_6 & 0 \\ 0 & N_6 \end{pmatrix} \begin{pmatrix} \hat{n}_x \\ \hat{n}_y \end{pmatrix} \|\mathbf{J}\| d\xi \quad (4.3.7.13)$$

```
using LinearAlgebra
```

```
function main()
```

```

    X = [
        4 0;
        4 4;
        3 2;
    ]

```

```

]

dNdx(x) = [
    x - 1/2;
    x + 1/2;
    -2 * x
]

# define quad_2 values
quad_2 = [
    (-1 / sqrt(3), 1.0),
    ( 1 / sqrt(3), 1.0)
]

# shape functions as vector
N(x) = [
    (x/2*(x-1)) 0;
    0 (x/2*(x-1));
    (x / 2 * (x + 1)) 0;
    0 (x / 2 * (x + 1));
    (1 - x^2) 0;
    0 (1 - x^2);
]

function normal(x)
    J = X' * dNdx(x)
    hatJ = J / norm(J)
    return [
        hatJ[2];
        -hatJ[1];
    ]
end

result = zeros(6)

for (x, w) in quad_2
    result += N(x) * normal(x) * norm(X' * dNdx(x)) * w
end

return result
end

main()

```

```

6-element Vector{Float64}:
 0.6666666666666667
 0.6666666666666662
 0.6666666666666665
-0.6666666666666662
 2.6666666666666665
 0.0

```

Thus

$$\mathbf{F}_t = \begin{pmatrix} F_{2tx} \\ F_{2ty} \\ F_{3tx} \\ F_{3ty} \\ F_{6tx} \\ F_{6ty} \end{pmatrix} = \frac{-q}{3} \begin{bmatrix} 2 \\ 2 \\ 2 \\ -2 \\ 8 \\ 0 \end{bmatrix} \quad (4.3.7.14)$$

13. Exercise 4.3.8

For the element below, determine the corresponding nodal forces if a normal traction $t_n = 1$ kPa is applied on the shaded face.

13.1 Solution

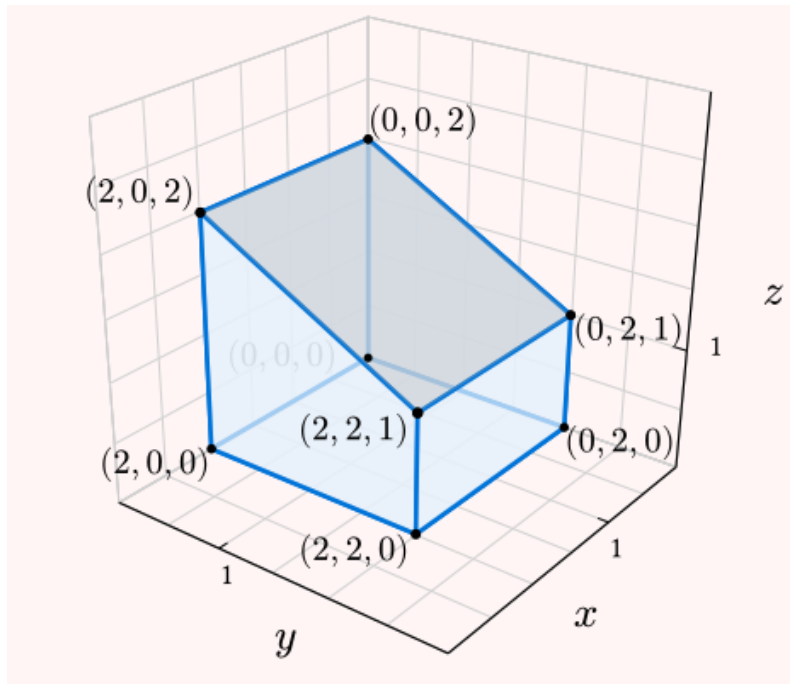


Figure 13.1: Element provided

Normal traction is defined as

$$\mathbf{F}_n = t_n \int_A \mathbf{N}^T \hat{\mathbf{n}} dA \quad (4.3.8.1)$$

The shape functions for a 4-node square 3D plane are

$$N_i = \frac{1}{4}(1 + \xi_i \xi)(1 + \eta_i \eta) \quad (4.3.8.2)$$

Thus, to solve this problem, we will use the following julia script

```
using LinearAlgebra

function main()
    tn = -1 # kPa

    X = [
        2 2 1;
```

```

    0 2 1;
    0 0 2;
    2 0 2;
]

# shape functions as vector
N(xi, eta) = [
    1/4 * (1 - xi) * (1 - eta);
    1/4 * (1 + xi) * (1 - eta);
    1/4 * (1 + xi) * (1 + eta);
    1/4 * (1 - xi) * (1 + eta);
]

# its derivatives
dNdx(xi, eta) = [
    (1/4 * (-1 + eta)) (1/4 * (-1 + xi));
    (1/4 * (1 - eta)) (1/4 * (-1 - xi));
    (1/4 * (1 + eta)) (1/4 * (1 + xi));
    (1/4 * (-1 - eta)) (1/4 * (1 - xi));
]

N_matrix(xi, eta) = transpose([
    N(xi, eta)[1] * Matrix(1I, 3,3)
    N(xi, eta)[2] * Matrix(1I, 3,3)
    N(xi, eta)[3] * Matrix(1I, 3,3)
    N(xi, eta)[4] * Matrix(1I, 3,3)
])

Jacobian(xi, eta) = X' * dNdx(xi, eta)

function normal(xi, eta)
    J = Jacobian(xi, eta)
    n = cross(J[:, 1], J[:, 2])
    return n / norm(n)
end

quad_3 = [
    (-sqrt(3/5), 5/9),
    (0, 8/9),
    (sqrt(3/5), 5/9),
]

# solving the integral
result = zeros(12)

for (xi, wi) in quad_3
    for (eta, wj) in quad_3
        J = Jacobian(xi, eta)
        result += N_matrix(xi, eta)' * normal(xi, eta) * sqrt(det(J' * J)) * wi
    * wj
    end
end

return result * tn
end

main()

```

```

12-element Vector{Float64}:
-0.0
-0.5

```

```

-1.0
-0.0
-0.5
-1.0
-0.0
-0.5
-1.0
-0.0
-0.5
-1.0

```

Thus, according to the local mapping (node 1 at (2, 2, 1))

$$\begin{pmatrix} F_{x_1} \\ F_{y_1} \\ F_{z_1} \\ F_{x_2} \\ F_{y_2} \\ F_{z_2} \\ F_{x_3} \\ F_{y_3} \\ F_{z_3} \\ F_{x_4} \\ F_{y_4} \\ F_{z_4} \end{pmatrix} = - \begin{pmatrix} 0.0 \\ 0.5 \\ 1.0 \\ 0.0 \\ 0.5 \\ 1.0 \\ 0.0 \\ 0.5 \\ 1.0 \\ 0.0 \\ 0.5 \\ 1.0 \end{pmatrix} \quad (4.3.8.3)$$